Yixuan Huang

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Google Scholar | Semantic Scholar | Lab

Education

Example University *Ann Arbor, MI*
Ph.D. in Mechanical Engineering *Expected May 2028*

Example Institute of Technology *Boston, MA*
B.S. in Electrical and Computer Engineering *May 2023*

Research Interests

Robotics; autonomous systems; learning-based control; robot perception; human-robot interaction; planning under uncertainty.

Publications

*(The superscript * indicates equal contribution.)*

Preprints

[P1] **Jordan Scholar***, Alex Researcher*, and Taylor Advisor. **Learning-Enabled Planning for Robust Robotic Manipulation**. *arXiv preprint*, 2026. [Preprint] [Code] [BibTeX]

[P2] **Jordan Scholar**, Casey Collaborator, and Taylor Advisor. **Multi-Robot Rearrangement Planning under Resource Constraints**. *Manuscript in preparation*, 2026.

Conference Papers

[C1] **Jordan Scholar**, Alex Researcher, and Taylor Advisor. **Planning for Multi-Arm Tabletop Rearrangement with Time and Buffer Optimization**. *Robotics: Science and Systems (RSS)*, 2026. Los Angeles, CA. [Preprint] [Code] [BibTeX]

[C2] **Jordan Scholar**, Morgan Engineer, and Taylor Advisor. **A Benchmark for Evaluating Perception-Aware Robot Planning**. *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. Vienna, Austria. [Project] [Data]

[C3] Alex Researcher, **Jordan Scholar**, and Taylor Advisor. **Efficient Buffer-Aware Planning for Tabletop Manipulation**. *Conference on Robot Learning (CoRL)*, 2025. Seoul, South Korea. [PDF] [Code]

Journal Articles

[J1] **Jordan Scholar** and Taylor Advisor. **Learning-Based Robotic Manipulation with Physical Interaction**. *IEEE Transactions on Robotics*, 2026. [DOI] [BibTeX]

Talks

Invited Talks

[T1] **Planning for Multi-Arm Rearrangement under Resource Constraints**. *Example Robotics Seminar, Example University*, Ann Arbor, MI. [Slides] [Video] *Mar. 2026*

[T2] **Learning-Enabled Planning for Robust Robotic Manipulation**. *Autonomous Systems Seminar*, Cambridge, MA. [Abstract] *Jan. 2026*

Research Experience

Example Robotics Laboratory *Ann Arbor, MI*
Graduate Research Assistant *Sep. 2023 – Present*
Topics: Robotic Manipulation, Planning, Learning-Based Control
Advisor: Prof. Taylor Advisor

- Developed planning and control pipelines for evaluating robotic manipulation under uncertainty.
- Designed simulation experiments to compare task success, execution time, and robustness across methods.

Example Autonomous Systems Group *Boston, MA*
Undergraduate Research Assistant *Jan. 2022 – May 2023*
Topics: Autonomous Navigation, Sensor Fusion, Field Robotics
Advisor: Prof. Casey Mentor

- Implemented data processing tools for analyzing robot trajectories and sensor logs.
- Assisted with field experiments involving mobile robots operating in semi-structured environments.

Teaching Experience

Instructor

Example University, **Introduction to Robotics**. [Course Website] [Syllabus] *Fall 2026*

Example University, **Robot Motion Planning**. *Winter 2026*

Teaching Assistant

Example University, **Computational Methods for Engineers**. [Course Website] *Spring 2025*

Example Institute of Technology, **Programming for Robotics**. *Fall 2024*

Guest Lecturer

Example Robotics Laboratory, **Planning under Uncertainty**. [Slides] *2026*

Selected Projects

Simulation Platform for Robot Rearrangement *2025*
Python, MuJoCo, Motion Planning

- Built a simulation environment for testing tabletop object rearrangement strategies.
- Evaluated methods using task completion time, collision rate, and number of intermediate placements.

Visual Perception Pipeline for Manipulation *2024*
Computer Vision, Depth Sensing, 3D Reconstruction

- Implemented a perception pipeline for estimating object geometry from RGB-D observations.
- Integrated reconstructed scene information into downstream robotic grasp and placement planning.

Learning-Based Control Benchmark *2024*
PyTorch, Reinforcement Learning, Control

- Compared model-free control methods on simulated continuous-control tasks.
- Analyzed policy performance across random seeds, reward designs, and environment perturbations.

Awards & Honors

Graduate Research Fellowship *2025*
Example University
Outstanding Teaching Assistant Award *2024*
Department of Mechanical Engineering, Example University

Dean's List *2020 – 2023*
Example Institute of Technology

Leadership & Service

Graduate Student Mentor *2024 – Present*
Example Robotics Laboratory

Reviewer *2025*
Example Student Research Symposium

Workshop Organizer *2024*
Robotics Reading Group, Example University

Media

Interviews

[M1] **Interview on Multi-Robot Planning**. *Example Robotics Podcast*. [Interview] *May 2026*

[M2] **Building Reliable Robots for Everyday Environments**. *Example Engineering Magazine*. [Article] *Feb. 2026*

Press Coverage

[M3] **Robots That Learn to Rearrange Cluttered Tables**. *Example Science News*. [Article] *Jun. 2026*

[M4] **New Benchmark Highlights Challenges in Robot Planning**. *Example University News*. *Dec. 2025*

Skills

Programming: Python, C++, Java, MATLAB

Tools: Linux, Git, LaTeX, Docker

Research Areas: Robotics, machine learning, data analysis